OG-PHL: PEP Simulation Results

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PEP Simulation Overview

- The Philippines Energy Plan (PEP) 2023 simulation models the economic and health impacts of energy sector transitions
- We compare a baseline scenario to the PEP 2023 scenario over a 50-year time horizon
- Key dimensions analyzed:
  - Energy sector parametrization (TFP, costs, investments)
  - Environmental outcomes (CO2e, PM2.5)
  - Health impacts (mortality rates, deaths averted)
  - Macroeconomic effects (GDP, capital, consumption, labor)
  - Fiscal implications (government spending, debt)
  - Distributional impacts (prices, consumption, income, labor supply)

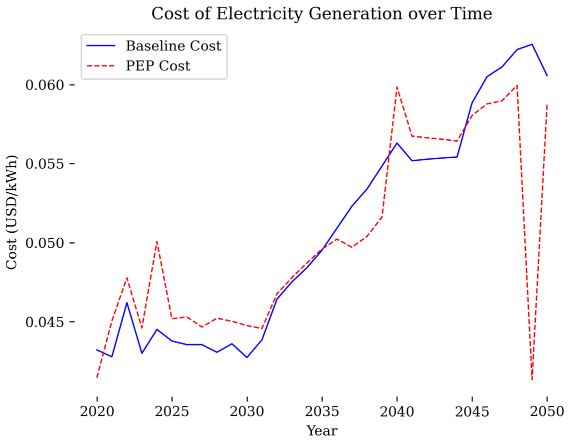
Input-Output Table for Multi-Industry Calibration

	Natural Resources	Electricity	Construction, Trade, Services	Manu.
Food	0.120	0.000	0.202	0.679
Energy & water	0.021	0.107	0.329	0.543
Non-durables	0.043	0.006	0.628	0.323
Durables	0.011	0.011	0.236	0.743
Services	0.041	0.003	0.772	0.185

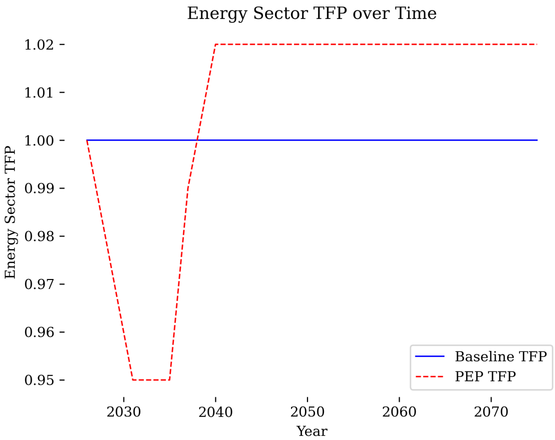
Consumption Category Shares for Multi-Industry Calibration

Consumption Category	Expenditure Share
Food	0.357
Energy and water	0.014
Non-durables	0.021
Durables	0.103
Services	0.505

CLEWs: Cost of Electricity Generation over Time



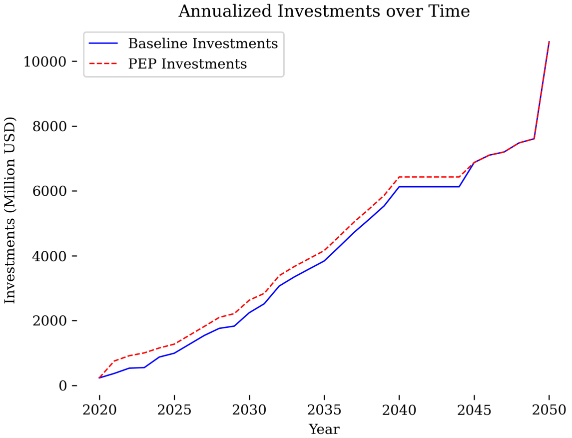
OG-PHL: Energy Sector TFP over Time



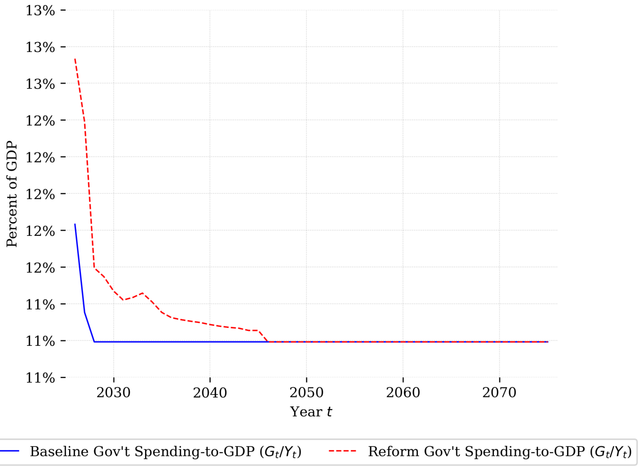
PEP23 Investments

- PEP23 estimates total investments of 19.6-27.8 trillion PHP over 2023-2050
- This includes both government and private investments
- Government investments (both direct and via incentives and partnerships) are about 10% of this total
- For our OG-PHL simulation, we assume government investments of 2.6 trillion PHP, spread over 27 years
- Private investment in endogenous in OG-PHL

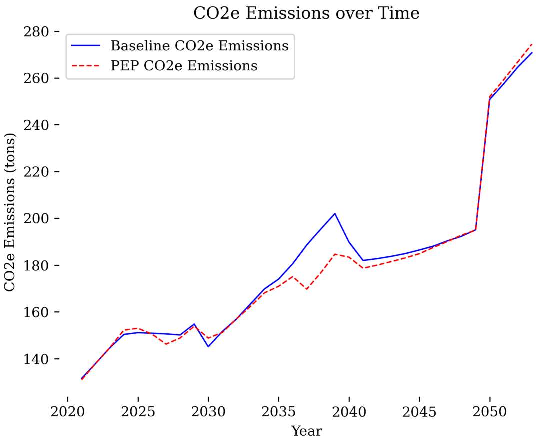
CLEWs: Annualized Investments over Time



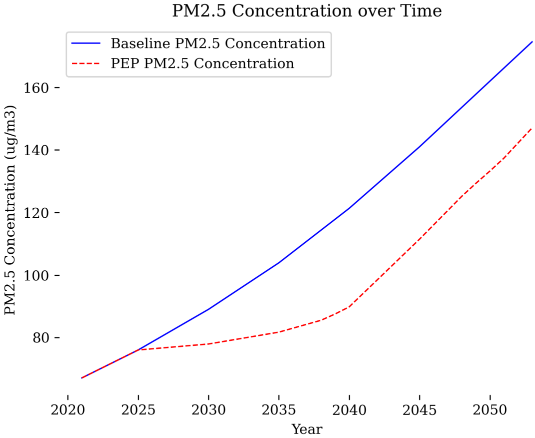
OG-PHL: Government Spending over Time



CLEWs: CO2 Emissions over Time



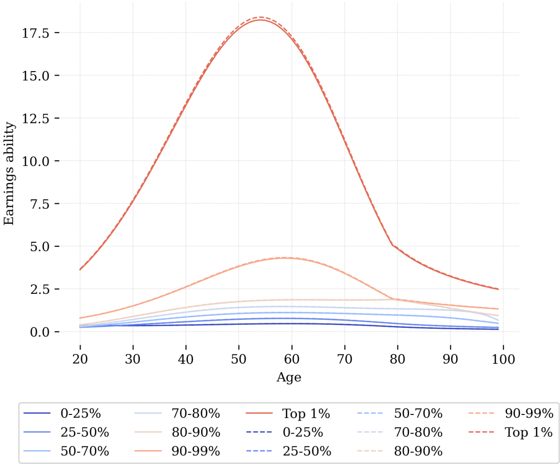
CLEWs: PM2.5 Concentration over Time



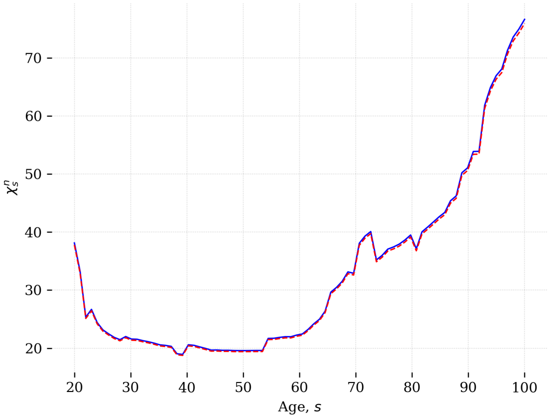
Mapping PM2.5 to Labor Productivity

- There are strong links in the literature between air pollution (PM2.5) and economic outcomes, including labor productivity
- He, Liu, and Salvo (*AEJ: Applied*, 2019) find that a **50% decline in PM2.5 reduces productivity by 3%**
- We use this mapping to adjust key productivity parameters in OG-PHL:
  - Ability profiles
  - Disutility of labor parameters
- CLEWS finds a 15% reduction in PM2.5 under PEP scenario, leading to a 0.9% increase in productivity

Ability Profiles in Baseline and PEP



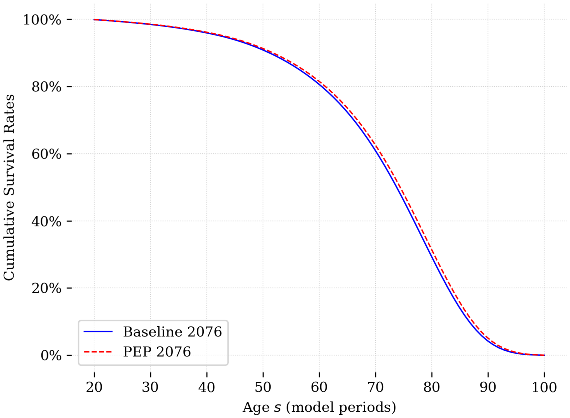
Disutility of Labor in Baseline and PEP



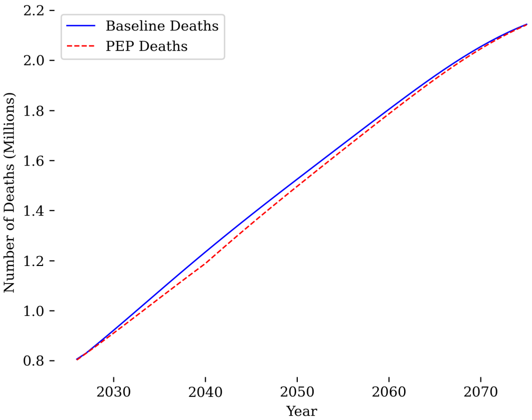
Mapping PM2.5 to Mortality

- A substantial scientific literature establishing a link between air pollution (PM2.5) and mortality:
  - Pope et al. (*Air Quality, Atmosphere, and Health*, 2017): 6-34% change in mortality per 10μg/m^3 change in PM2.5
  - Kim et al. (*Environmental Health*, 2021): 10μg/m^3 increase in in PM2.5 lowers life expectancy by 0.7-0.9 years
  - Puett et al. (*Environmental Health Perspectives*, 2009): 10μg/m^3 increase in PM2.5 leads to 4% increase in all-cause mortality risk
- We take the estimates of Pope et al. (*Air Quality, Atmosphere, and Health*, 2017) to map PM2.5 concentrations to mortality risk
- The decrease in PM2.5 under the PEP scenario is about 10μg/m^3, which we map to a conservative 4% decrease in mortality risk

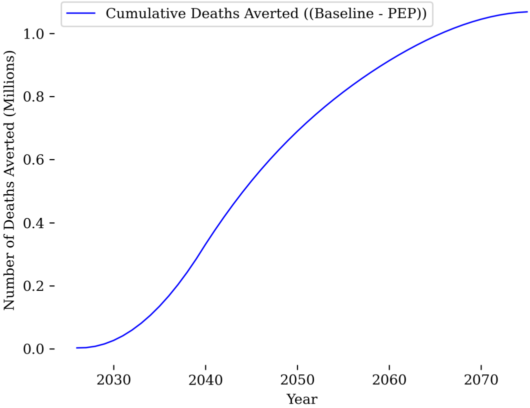
Mortality Baseline and PEP



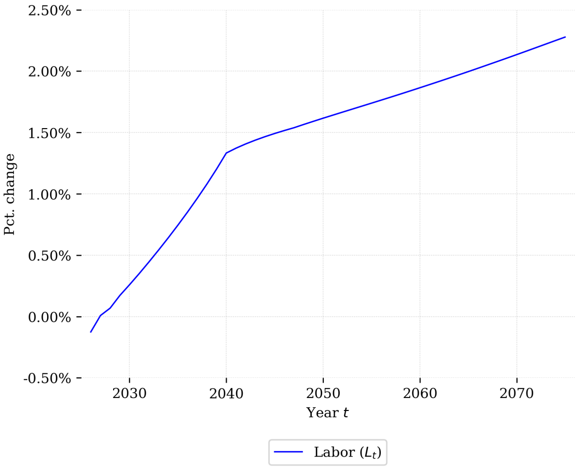
Annual Deaths over Time



Cumulative Deaths Averted over Time

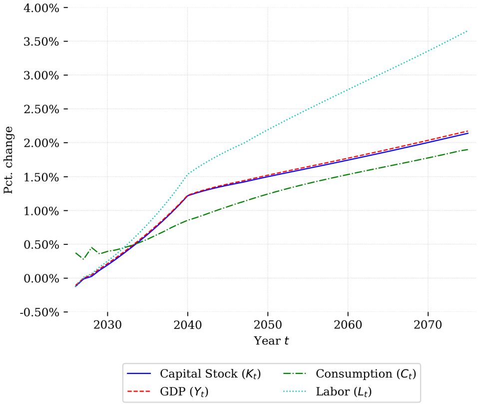


Labor Supply over Time



Aggregate Variables over Time

- Percentage changes in capital (K), output (Y), consumption (C), and labor (L)

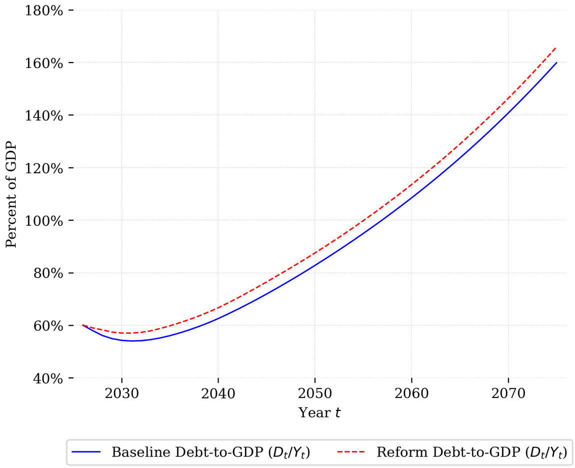


Long Run Effects on GDP

Discount Rate	NPV of GDP Increase (Trillions of PHP)
1%	5,073.030
2%	2,216.387
3%	998.952
4%	466.805
5%	227.508

Note: PEP23 Estimate of costs: 26-42 Trillion PHP

Debt-to-GDP Ratio over Time

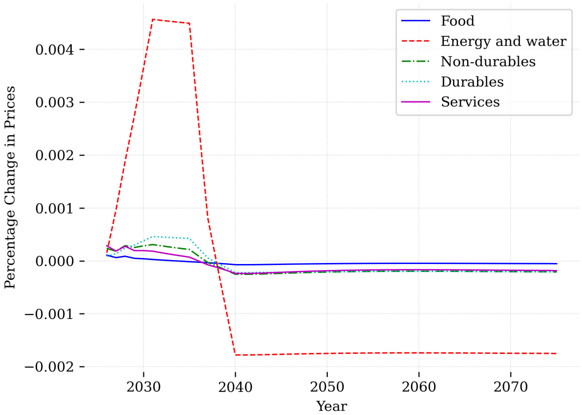


Long run revenue effects

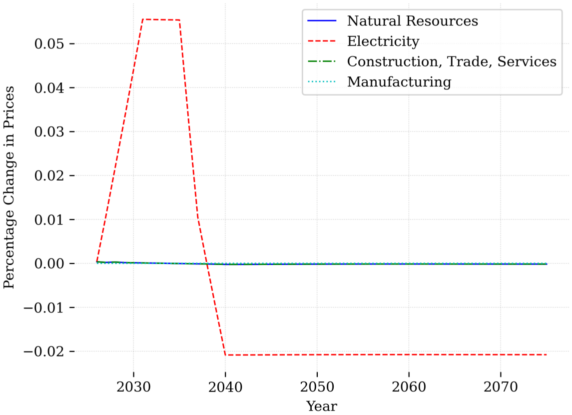
Discount Rate e	NPV of Tax Revenue Increase (Trillions PHP)
1%	294.667
2%	131.213
3%	60.710
4%	29.408
5%	15.041

Note: PEP23 Estimate of government costs: 2.6-4.2 Trillion PHP

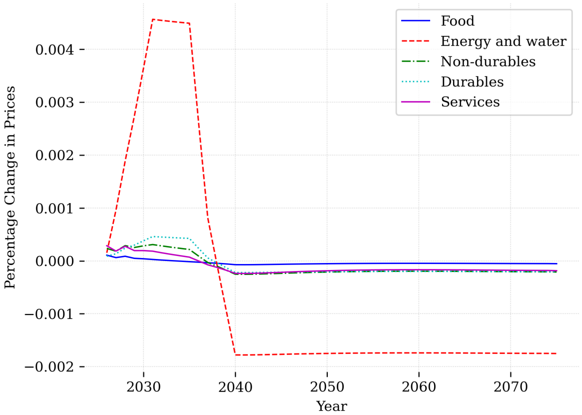
Percentage Change in Consumption Good Prices



Percentage Change in Production Good Prices

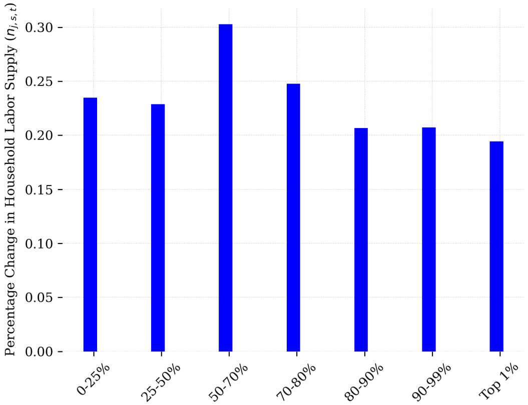


Percentage Change in Consumption Good Prices



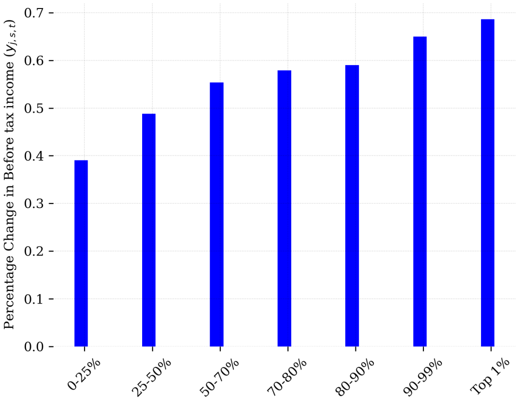
Percentage Change in Labor Supply by Ability Group

- Changes over 2036-2046



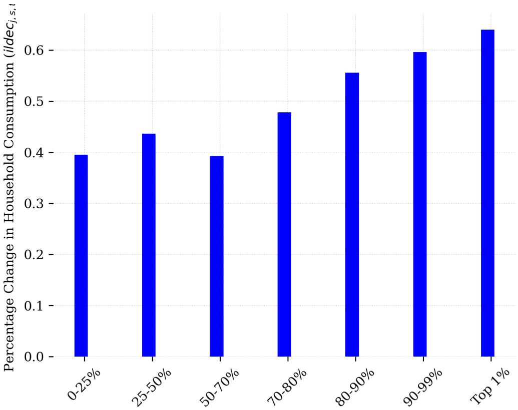
Percentage Change in Income by Ability Group

- Changes over 2036-2046



Percentage Change in Consumption by Ability Group

- Changes over 2036-2046



Key Findings

- Energy Transition:
  - PEP scenario shows increased energy sector TFP over time
  - Lower electricity generation costs in PEP scenario
  - Significant upfront investments required
- Environmental Benefits:
  - Reduced CO2e emissions
  - Lower PM2.5 concentrations
- Health Impacts:
  - Improved survival rates and lower mortality
  - Substantial cumulative deaths averted
  - Increased labor supply due to better health

Key Findings (continued)

- Macroeconomic Effects:
  - Positive impacts on capital, output, consumption, and labor
- Fiscal Implications:
  - Government spending increases for energy investments
  - Changes in debt-to-GDP ratio
- Distributional Effects:
  - Changes in relative prices across sectors
  - Electricity prices adjust over time
  - Differential impacts across ability groups